

ABOUT ME

Having had a foot in the engineering industry since the summer between my freshman and sophomore years of college, I am looking to transition out of full time teaching back to full time industry work. I would like to join a long-term team that I can be a part of to push the company to new levels of success while growing in my career. I have designed products ranging from medical monitoring devices to teaching aids to drones for precision agriculture, working in three countries for customers both local and remote, some of whom I have never met in person. I have also collected, analyzed, and reported on data for governmental reports, helped run a business based on data collection and reporting, managed employees and research projects, and designed many printed circuit boards. I have 18 years of industry experience and 14 years of teaching experience with many years of doing both. Teaching has given me years of experience understanding, debugging, and suggesting refactoring of code.

EDUCATION

Bachelor of Science in Engineering	<i>Emphasis in Electrical Engineering</i>	2001
Master of Science	<i>Electrical and Computer Engineering</i>	2004
Doctor of Philosophy	<i>Electrical and Computer Engineering</i>	2009

RECENT EMPLOYMENT

Hardware and Software Consultant 2011 – Present
ANITRACKS, LLC

- Parsed, processed, stored, and released data for docketreminder.com and research projects
- Developed algorithms for UAV detection of maritime non-ocean objects in still images
- Developed system and algorithms for automatic video stream quantization of bed bug movement
- Implemented RFID infrastructure for monitoring butterflies

Associate Professor in Computer Engineering 2021 – 2025

EMBRY-RIDDLE AERONAUTICAL UNIVERSITY, PRESCOTT

- Created new course, wrote textbook, lab manual, and designed lab robot
- Directed and managed embedded machine learning research group
- Received the College of Engineering Teaching Innovation Award 2021-2022
- Supervised, advised, and managed 8 undergraduate student projects

Associate Professor & Program Manager for Computer Science & Math 2020-2021

UNION ADVENTIST UNIVERSITY, LINCOLN, NE

- Managed corporately sponsored research using drones for precision agriculture and mRNA extraction
- Redesigned the Computing degree
- Managed the computer science and mathematics degree programs
- Scheduled courses
- Searched for and hired adjunct professors
- Taught computer science courses
- Supervised, advised, and managed 3 undergraduate student projects

Assistant Professor 2014-2020

UNION ADVENTIST UNIVERSITY, LINCOLN, NE

- Managed the engineering program
- Taught engineering and computer science classes
- Chaired the Campus Technology Committee for 2 years
- Worked on an institutional research data team to improve Union
- Managed corporately sponsored research involving drones and mRNA

Applications Engineer

2013 – 2014

UNIVERSITY OF NEBRASKA-LINCOLN

Department of Computer Science & Engineering

- Interfaced between basic research and field applications for multi-rotor drones
- Searched for and applied to novel funding sources
- Helped to manage graduate research projects

Electrical Engineer

2012-2013

UNITED STATES DEPARTMENT OF AGRICULTURE, AGRICULTURAL RESEARCH SERVICE, CENTER FOR MEDICAL, AGRICULTURAL & VETERINARY ENTOMOLOGY

- Worked towards commercial prototype of active acoustic agricultural insect pest trap
- Managed undergraduate interns and employees
- Implemented automatic recognition of insect calls using an 8-bit microcontroller

Director of Engineering Development

2009-2011

TETRACAM, INC.

- Directed engineering, managed projects and engineering group, developed products, wrote grants, and handled marketing, sales, and customer service in the fields of precision agriculture, hunting cameras, and UAVs.

CODE REPOSITORIES

Academic GitHub: <https://github.com/semcneil>

Personal GitHub: <https://github.com/anitracks>

Personal GitLab: https://gitlab.com/anitracks_union/

SKILLS

Python, C, C++, git, Raspberry Pi, Arduino, RP2040, ESP8266, ESP32, Linux, MATLAB, Altium, MSP430s, R statistical computing, SolidWorks, Bruker spectrometers, TCL, RF circuit assembly, OpenCV, Mentor Graphics, Mathematica, PHP, MySQL, AutoCAD, Excel, Word, PowerPoint, metal lathe, metal mill, hand tools